## Normalization

Violation form : 1NF

The seat_numbers column contains composite values.In 1FN each row have unique identifier

1NF: There are two seats in the booking, so we are creating two separate lines to ensure there is one seat per cell.Table: Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price).Primary key: booking_id, passenger_passport_number

2NF:Table 1: Passengers(passenger_passport_number, passenger_full_name)

Primary key: passenger_passport_number

Table 2: Bookings(booking_id, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)

Primary key : booking_id

Table 3: Booking_passengers(booking_id, passenger_passport_number, seat_number)

Primary key: passenger_passport_number, booking_id

3NF:

Table 1: Passengers(passenger_passport_number, passenger_full_name)

Primary key: passenger_passport_number

Table 2: Airports(airport_name, city)

PK : airport_name

Table 3: Flights(flight_number, departure_airport_name, arrival_airport_name)

PK: flight_number

FK: departure_airport_name, arrival_airport_name

Table 4:

Bookings(booking_id, , flight_number, ticket_price)

PK: booking_id

Fk: flight_number

Table 5: Booking_passengers(booking_id, passenger_passport_number, , ticket_price)

PK: booking_id, passenger_passport_number

FK: booking_id,passenger_passport_number

- booking_id -> flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price

- passenger_passport_number -> passenger_full_name

- flight_number -> departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name

- departure_airport_name -> departure_city

- arrival_airport_name -> arrival_city

Does table Fligths satisfied 3NF ?

Yes, all foreign key attributes are only ID references to external entities (flight_number, departure_airport_name, arrival_airport_name)

| passenger_passport_number | passenger_full_name |
|---|---|
| N112233 | Anna Sidorova |
| N123456 | Jamilya Zholdas |
| N456789 | Petr Petrov |
| N654321 | Ayazhan Kydyr |
| N987654 | Ivan Ivanov |

| airport_name | city |
|---|---|
| Almaty International Airport | Almaty |
| Astana International Airport | Astana |
| Dubai International Airport | Dubai |
| Heathrow Airport | London |
| JFK Airport | New York |

| flight_number | departure_airport_name | arrival_airport_name |
|---|---|---|
| BA123 | Heathrow Airport | JFK Airport |
| KC871 | Almaty International Airport | Astana International Airport |
| KC872 | Astana International Airport | Almaty International Airport |
| KC901 | Almaty International Airport | Dubai International Airport |
| UA456 | JFK Airport | Heathrow Airport |

| booking_id | flight_number | ticket_price |
|---|---|---|
| 101 | KC871 | 150.0 |
| 102 | KC901 | 350.0 |
| 103 | KC872 | 160.0 |
| 104 | BA123 | 750.0 |
| 105 | UA456 | 700.0 |

| booking_id | passenger_passport_number | seat_number |
|---|---|---|
| 101 | N123456 | 12A |
| 101 | N654321 | 12B |
| 102 | N987654 | 15C |
| 103 | N456789 | 01A |
| 104 | N112233 | 22D |

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## DDL:

Primary key is their ID

Query #1 Execution time: 18.72ms

| airline_id | airline_code | airline_name | airline_country | created_at | updated_at | info |
|---|---|---|---|---|---|---|

There are no results to be displayed.

Query #2 Execution time: 1.09ms

| airport_id | airport_name | country | state | city | created_at | updated_at |
|---|---|---|---|---|---|---|

There are no results to be displayed.

Query #3 Execution time: 0.94ms

| passenger_id | first_name | last_name | date_of_birth | gender | country_of_citizenship | country_of_residence | passport_number | created_at | updated_at |
|---|---|---|---|---|---|---|---|---|---|

There are no results to be displayed.

Query #4 Execution time: 0.8ms

| flight_id | sch_departure_time | sch_arrival_time | departing_airport_id | arriving_airport_id | departing_gate | arriving_gate | airline_id | act_departure_time | act_arrival_time | cr |
|---|---|---|---|---|---|---|---|---|---|---|

There are no results to be displayed.

Query #5 Execution time: 0.85ms

| booking_id | flight_id | passenger_id | booking_platform | created_at | updated_at | status | price |
|---|---|---|---|---|---|---|---|

There are no results to be displayed.

Query #6 Execution time: 0.84ms

| baggage_check_id | check_result | created_at | updated_at | booking_id | passenger_id |
|---|---|---|---|---|---|

There are no results to be displayed.

Query #7 **Execution time: 0.69ms**

| baggage_id | weight_in_kg | created_at | updated_at | booking_id |
|---|---|---|---|---|

There are no results to be displayed.

Query #8 **Execution time: 0.61ms**

| boarding_pass_id | booking_id | seat | boarding_time | created_at | updated_at |
|---|---|---|---|---|---|

There are no results to be displayed.

Query #9 **Execution time: 6.54ms**

| booking_flight_id | booking_id | flight_id | created_at | updated_at |
|---|---|---|---|---|

There are no results to be displayed.

Query #10 **Execution time: 1.08ms**

| security_check_id | check_result | created_at | updated_at | passenger_id |
|---|---|---|---|---|

There are no results to be displayed.

```
CREATE TABLE Airline_info (
    airline_id INT PRIMARY KEY,
    airline_code VARCHAR(30) NOT NULL,
    airline_name VARCHAR(50) NOT NULL,
    airline_country VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    info VARCHAR(50) NOT NULL
);

CREATE TABLE Airport (
```

Text to DDL

```
RENAME TABLE Airline_info TO Airline;

ALTER TABLE Booking RENAME COLUMN price TO ticket_price;

ALTER TABLE Flights MODIFY COLUMN departing_gate TEXT;

ALTER TABLE Airline DROP COLUMN info;

ALTER TABLE Security_check
    ADD CONSTRAINT fk_security_check_passengers
    FOREIGN KEY (passenger_id) REFERENCES
Passengers(passenger_id);
```

Results

Copy as Markdown

There are no results to be displayed.

Query #8 **Execution time: 22.31ms**

There are no results to be displayed.

Query #9 **Execution time: 20.53ms**

There are no results to be displayed.

Query #10 **Execution time: 21.2ms**

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